

DWH_staging (Synapse) vs daily_snapshot (DBX) — full audit

Generated 2026-05-24 09:11 UTC • source: tools/_compare_staging_vs_snapshot.py • detail: tools/_staging_vs_snapshot_diff.csv (139 rows)

Scoreboard

verdict	count
MATCH (row counts identical, freshness identical where checked)	131
SYN_ONLY_MISSING (table exists only in DBX)	4
DBX_ONLY_MISSING (table exists only in Synapse)	3
DBX_SHORT (DBX < 50% of Synapse rows)	1

131 / 135 common tables are bit-for-bit identical in row count, and where an update-timestamp column exists they also match to the millisecond. The migration's daily snapshot layer is in **very good shape overall**; the failures are localized.

CRITICAL — drives observed Dim_Customer divergences

1. customerfinancedb_customer_firsttimedeposits is a VIEW pointing at the wrong source

- DBX has TWO objects with the same data signature:
 - **Table** dwh_daily_process.daily_snapshot.customerfinancedb_customer_globalftds = **1,074 rows** (min Gcid 1,663,695 / max 48,378,600 / sum_gcid 43,304,142,235)
 - **View**
dwh_daily_process.daily_snapshot.customerfinancedb_customer_firsttimedeposits = **5,566,497 rows** — body is `SELECT * FROM main.bi_db.bronze_customerfinancedb_customer_firsttimedeposits` (full history bronze)
- Synapse equivalent DWH_staging.CustomerFinanceDB_Customer_FirstTimeDeposits = **1,074 rows** (signature **identical** to the DBX table).

So the proper snapshot exists and matches Synapse exactly, but the view shadows it with the historical bronze table. `sp_dim_customer_dl_to_synapse` reads `...daily_snapshot.CustomerFinanceDB_Customer_FirstTimeDeposits` — hits the view — and dumps 5.5M rows of historical FTDs into `Ext_CustomerFinanceDB_Customer_FirstTimeDeposits` (vs 1,074 in Synapse).

This is the root cause of the `FTDPlatformID` reassignments (–110k TradingPlatform / +89k eMoney) and the `IsDepositor` shortfall we observed in `Dim_Customer`.

Fix (one-liner):

```
CREATE OR REPLACE VIEW dwh_daily_process.daily_snapshot.customerfinancedb_customer_first
AS SELECT * FROM dwh_daily_process.daily_snapshot.customerfinancedb_customer_globalfts;
```

or drop the view and rename the table. Either way, no SP changes are needed.

2. sts_audit_useroperationsdata snapshot is 1 day deep, Synapse is 8 days deep

side	rows	min(CreatedAt)	max(CreatedAt)	s
Synapse DWH_staging.STS_Audit_UserOperationsData	53,075,981	2026-05-16 00:00	2026-05-23 23:59:59	8 c
DBX daily_snapshot.sts_audit_useroperationsdata	6,879,309	2026-05-23 00:00	2026-05-23 23:59:59	c c (f

The 2FA INSERT does `ROW_NUMBER() OVER (PARTITION BY Gcid ORDER BY CreatedAt DESC) ...` WHERE `LoginTypeName` in ('TwoFactor_UpdatedByUser', 'TwoFactor_UpdatedByAdmin'). If a user's last 2FA toggle happened before 2026-05-23, their state is invisible in DBX → directly produces the –87% shortfall in `Ext_Dim_Customer_2FA` (33,586 in Synapse vs 4,320 in DBX).

This is the second root cause of the `Dim_Customer` divergences.

Fix: extend the `daily_snapshot` retention for this table from 1 day to the same window Synapse keeps (looks like rolling 7-day or longer). Note that the SP's "latest 2FA toggle per `Gcid`" semantics arguably want **lifetime** retention (or a pre-aggregated current-state view).

HIGH — entirely missing tables in DBX

3. IP2Location — 5,311,956 rows in Synapse, missing in DBX

No object with this name exists anywhere in `main.*` or `dwh_daily_process.*`. If any SP joins to it (typical use: IP → country/region lookup), that path is silently broken.

4. PriceLog_History_CurrencyPrice_Active_5_days — 632,245,921 rows in Synapse, missing in DBX

The 1-day variant `PriceLog_History_CurrencyPrice_Active` (62M rows) is present and matches exactly, but the 5-day rolling variant was never migrated. If any mark-to-market / EOD pricing job depends on 5-day rolling pricing, it will fail or fall back to 1-day.

INFO — Synapse-only tables (exist in DBX but not in Synapse

DWH_staging)

These do not point to drift, they're just objects that were either renamed on migration or added later in DBX without a matching staging table:

DBX-only table	DBX rows	note
customerfinancedb_customer_globalftds	1,074	renamed version of the FTD table — see finding #1
etoro_backoffice_campaign	15,571	new table, no Synapse equivalent in DWH_staging
fiktivo_dbo_tblaff_affiliatesgroups	243	likely overlap with the existing fiktivo_affiliateadmin_affiliatesgroups (248 rows) — verify before relying on either
userapidb_dictionary_tanganystatus	6	small dictionary table, likely fine

131 MATCH tables — what "MATCH" means

Row count is identical to the row in Synapse, and where the script could find a recognized update-timestamp column (UpdateDate / ValidFrom / Occurred / DateAdded / ModifiedDate / LastUpdateDate / CreatedAt / Updated), the MAX timestamp also matched to the millisecond.

Largest matched tables sanity-check well:

table	rows
etoro_history_backofficecustomer	612,031,423
etoro_trade_openpositionendofday	138,584,405
etoro_history_withdrawtofundingaction	144,631,847
etoro_history_withdrawaction	102,137,787
etoro_trade_positionshedgeserverchangelog	84,651,136
etoro_history_mirror	70,616,948
pricelog_history_currencyprice_active	62,007,759
fiktivo_affiliatecommission_registrationcommission	50,346,420
etoro_customer_customer / etoro_customer_customerstatic / etoro_backoffice_customer	47,867,332
fiktivo_affiliatecommission_registration	47,444,166
etoro_backoffice_customerdocument	44,065,455
etoro_backoffice_customerdocumenttodocumenttype	41,629,657
etoro_backoffice_customeralltimeaggregateddata	40,776,256

fiktivo_affiliatecommission_credit	40,513,062
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Caveat: row count + max timestamp match does not prove row-level content match. Anything driven by a WHERE ... filter that's wrong on either side could still diverge. If a specific Dim/Fact downstream looks suspicious, do a hash/aggregate spot-check on the relevant columns (as we did for FTD: min/max/sum GCID).

Recommended next actions

1. **Repoint the FTD view** (1-line SQL fix) — fixes the Dim_Customer.IsDepositor and FTDPlatformID skew immediately.
2. **Backfill sts_audit_useroperationsdata** with the full Synapse window (or change snapshot retention to ≥ 7 days, ideally lifetime for 2FA semantics) — fixes the 2FA shortfall.
3. **Decide on IP2Location and PriceLog_History_CurrencyPrice_Active_5_days** — either migrate them or audit which SPs/views still reference them and rewrite.
4. **Audit dwh_daily_process.daily_snapshot.* for other VIEW masks** — only one was found this round (customerfinancedb_customer_firsttimedeposits), but it's worth a one-time scan whenever new objects land.